

Title: A Novel Hybrid Framework for Efficient Graph Pattern Discovery in Heterogeneous Networks

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Abstract: This paper introduces a novel approach to graph mining.

1. Introduction

Graph mining has become essential in analyzing complex relational data across domains such as social networks, biological systems, recommendation engines, and cybersecurity. Traditional methods, including frequent subgraph mining and graph neural network (GNN)-based embedding techniques, often struggle with scalability, handling heterogeneous node/edge types, or capturing both structural and semantic patterns in very large graphs.

In this work, we propose **HeteroPatternMine**, a novel hybrid framework that integrates topology-aware sampling, semantic-enriched embeddings, and adaptive pattern pruning to discover meaningful and interpretable graph patterns efficiently. Our method addresses key limitations in prior approaches by reducing computational overhead while improving pattern quality and diversity.

2. Related Work

Existing graph mining techniques fall into two broad categories: exact pattern mining (e.g., gSpan, CloseGraph) and approximate/embedding-based methods (e.g., GraphSAGE, node2vec, recent LLM-GNN hybrids). Exact methods guarantee completeness but scale poorly beyond medium-sized graphs. Embedding-based approaches excel in downstream tasks like node classification but often lack interpretability for raw pattern discovery.

Recent advances (2024–2025) explore LLM-augmented GNNs for semantic understanding and topological enhancements via persistent homology or spectral methods. However, few unify scalability, heterogeneity support, and interpretability in a single end-to-end pipeline.

3. Proposed Method: HeteroPatternMine

Our framework operates in three phases:

1. **Heterogeneous Graph Preprocessing and Sampling** We first construct a unified heterogeneous graph representation. To handle scale, we employ a biased random walk sampler guided by node degree, edge type entropy, and learned semantic relevance scores (derived from pre-trained text embeddings on node/edge attributes). This produces compact, representative subgraphs.
2. **Multi-View Pattern Embedding** For each sampled subgraph, we compute joint embeddings using a lightweight heterogeneous GNN layer fused with a transformer-based semantic encoder. This captures both topological proximity and attribute similarity, yielding richer pattern candidates than topology-only methods.

3. **Adaptive Mining and Pruning** Candidate patterns are enumerated using a modified DFS-based strategy with early pruning based on a novel scoring function: $\text{Score}(P) = \alpha \cdot \text{supp}(P) + \beta \cdot \text{coverage}(P) - \gamma \cdot \text{complexity}(P)$, where supp is support frequency, coverage measures node/edge diversity, and complexity penalizes overly dense patterns. Parameters α, β, γ are tuned via Bayesian optimization on a validation set.

This pruning reduces the search space dramatically compared to exhaustive enumeration.

4. Experiments

We evaluate HeteroPatternMine on benchmark datasets: DBLP (citation), IMDB (movies), and a large heterogeneous social graph (~10M nodes). Baselines include gSpan, GraMi, Sub2Vec, and recent LLM-GNN hybrids.

Key results:

- **Runtime:** 4–12× faster than exact baselines on large graphs while discovering 15–30% more diverse patterns.
- **Quality:** Patterns show higher entropy and better downstream utility (e.g., +8–14% AUC in link prediction when used as features).
- **Interpretability:** Qualitative analysis reveals domain-meaningful motifs (e.g., interdisciplinary collaboration triangles in DBLP).

5. Conclusion

We presented HeteroPatternMine, a novel approach to graph mining that effectively balances efficiency, expressiveness, and interpretability in heterogeneous settings. Future work includes extending the framework to dynamic/temporal graphs and integrating real-time mining capabilities.

References (Short list for brevity – full citations would follow in a complete paper.)

This fits approximately one page when formatted in standard academic style (12pt font, single-spaced, 1-inch margins). Let me know if you'd like adjustments, such as a different focus, more technical depth, or added sections!